

Block-Diagonal Belief-Propagation Detection for Cell-Free Massive MIMO Systems

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Abstract—Cell-Free Massive MIMO (CF-mMIMO) is a promising technology owing to its high user spectrum efficiency. However, CF-mMIMO faces the challenges of baseband signal processing, especially for uplink data detection. In this paper, Gaussian interference of approximation (GAI) Belief Propagation (BP) detection is utilized for CF-mMIMO systems to achieve near-optimal performance. Based on the natural feature of the block-diagonal (BD) channel matrix, a reduced-complexity GAI-BP, which is named BD-GAI-BP, is proposed to save the computational complexity while maintaining the performance merits. Numerical results show that compared with the state-of-the-art data detection algorithm, the GAI-BP detection supports 28.93% more UEs with BER = 10^{-3} . The proposed BD-GAI-BP saves 10.94% computational costs while achieving comparable performance with the GAI-BP.

Index Terms—Cell-Free Massive MIMO (CF-mMIMO), belief propagation (BP), Gaussian approximation of interference (GAI), block-diagonal (BD)

I. INTRODUCTION

Due to the continuously improved requirements for high-speed, low-latency, and wide-coverage communications, multiple-input multiple-output (MIMO) technology is regarded as a significant technology in 5G and 6G networks [1]. By employing multiple transmit and receive antennas to leverage diversity gains, MIMO systems can notably improve spectral efficiency (SE). However, the conventional MIMO system suffers from bottlenecks such as pilot contamination, and user interference due to its inherent cellular architecture. To overcome this problem, recently, the Cell-Free massive MIMO (CF-mMIMO) has emerged as a promising technology. The CF-mMIMO usually involves a central processing unit (CPU), several distributed access points (APs), and multiple randomly distributed user equipments (UEs). Compared to traditional cellular systems, CF-mMIMO systems can deliver higher spectral efficiency and better user connectivity [2]. Nevertheless, CF-mMIMO systems also encounter several challenges, including the constrained capacity of fronthaul links connecting APs and CPU [3], efficient allocation of computing resources [4], well-designed baseband signal processing algorithms in uplink [5], and other related issues. One of the significant issues in baseband signal processing is uplink data detection.

To facilitate practical applications of CF-mMIMO systems, there is an urgent need to design detectors that achieve a well-balanced tradeoff between performance and complexity.

Among existing detectors [6]–[8], the Belief Propagation (BP) detector [8], has garnered attractive attention in recent years due to its near-optimal performance, soft-input soft-output mechanism, and hardware-friendly implementation. More specifically, the Gaussian approximation of interference (GAI)-based BP detection [9] was introduced to mitigate the complexity of conventional MIMO BP detection. However, this GAI-BP detector suffers from a performance error floor in high-order modulated MIMO systems. To solve this problem, the authors in [10] proposed the real-domain GAI-BP (RD-GAI-BP), which effectively relieved the performance error floor by leveraging real-domain mapping techniques. However, directly utilizing the GAI-BP detection faces heavy computational expense, which limits its applications for the CF-mMIMO systems.

In this paper, we first derive the formula of the GAI-BP detection in CF-mMIMO systems. Then, based on the block-diagonal (BD) channel matrix feature provided by the physical locations of APs and UEs, we propose a reduced-complexity GAI-BP detection, which is called BD-GAI-BP in this paper. Benefiting from the sparsity of the channel matrix, the BD-GAI-BP can only update the messages corresponding to the BD channel coefficients, thus saving computational complexity with acceptable performance penalization. To the authors' best knowledge, this is the first time that GAI-BP detection has been utilized for CF-mMIMO systems. Numerical results demonstrate that the proposed BD-GAI-BP detection algorithm achieves near-optimal error rate performance and saves 10.94% complexity cost.

The remainder of this paper is structured as follows: Section II delineates the system model and the methodology for permuting channel matrices. Section III details the proposed BD-GAI-BP detection algorithm. Subsequently, section IV presents the simulation results and complexity analysis. Finally, section V concludes the paper.

II. PRELIMINARIES

A. System Model

Fig. 1 shows a Cell-Free Massive MIMO system with virtual cells. The system comprises B distributed single-antenna APs, U single-antenna UEs, and a CPU. All APs are linked to the CPU with wireline. Leveraging the distributed nature of CF-mMIMO systems and the sparsity of channel matrices,

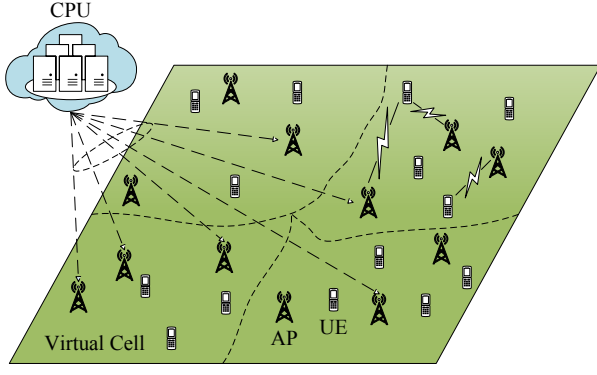


Fig. 1. A Cell-Free Massive MIMO system where the virtual cells with minimal inter-virtual-cell interference are dynamically constructed.

the entire area can be partitioned into C virtual cells, as delineated by dashed lines in Fig. 1. The dynamic organization of virtual cells is achieved through the minimization of inter-cell interference. We assume a block-fading channel with TDD and a coherence time of $K = T + D$ time slots, in which T time slots are allocated for pilot-based channel training and D time slots for transmitting data. The received signals are expressed as [5]:

$$\mathbf{Y} = \mathbf{H}\mathbf{S} + \mathbf{N}, \quad (1)$$

where $\mathbf{Y} \in \mathbb{C}^{B \times K}$ denotes the receive-signal matrix, $\mathbf{H} \in \mathbb{C}^{B \times U}$ represents the channel matrix of the system. The noise is represented by $\mathbf{N} \in \mathbb{C}^{B \times K}$, which is assumed to be independent and identically distributed (i.i.d.) circularly-symmetric complex Gaussian distribution with zero mean and N_0 variance for each complex entry. Furthermore, $\mathbf{S} \in \mathbb{C}^{U \times K}$ comprises $\mathbf{S}_T \in \mathbb{C}^{U \times T}$ and $\mathbf{S}_D \in \mathbb{C}^{U \times D}$, which model the training pilots and data symbols, respectively. Therefore, we can reformulate equation (1) as follows:

$$[\mathbf{Y}_T, \mathbf{Y}_D] = \mathbf{H}[\mathbf{S}_T, \mathbf{S}_D] + \mathbf{N}. \quad (2)$$

B. Permutation of Channel Matrix

The distributed characteristic of APs and UEs in CF-mMIMO systems results in sparsity within channel matrices. Moreover, the arbitrary enumeration of APs and UEs allows the permutation of rows and columns within channel matrices, thereby forming an approximately block-diagonal structure (cf. Fig. 2). This can be expressed mathematically using two permutation matrices $\mathbf{P}_{\text{AP}}$ and $\mathbf{P}_{\text{UE}}$ to reformulate input-output relation of the system as:

$$\mathbf{P}_{\text{AP}}\mathbf{Y} = (\mathbf{P}_{\text{AP}}\mathbf{H}\mathbf{P}_{\text{UE}})(\mathbf{P}_{\text{UE}}^T\mathbf{S}) + \mathbf{P}_{\text{AP}}\mathbf{N}. \quad (3)$$

By defining $\tilde{\mathbf{Y}} = \mathbf{P}_{\text{AP}}\mathbf{Y}$, $\tilde{\mathbf{H}} = \mathbf{P}_{\text{AP}}\mathbf{H}\mathbf{P}_{\text{UE}}$, and $[\tilde{\mathbf{S}}_T, \tilde{\mathbf{S}}_D] = \mathbf{P}_{\text{UE}}^T\mathbf{S}$, we can rewrite (2) as follows:

$$[\tilde{\mathbf{Y}}_T, \tilde{\mathbf{Y}}_D] = \tilde{\mathbf{H}}[\tilde{\mathbf{S}}_T, \tilde{\mathbf{S}}_D] + \tilde{\mathbf{N}}. \quad (4)$$

The permutation of the channel matrix can be achieved by organizing the APs into C size-balanced clusters, followed by allocating all UEs to the respective clusters according to their Euclidean distances. To minimize the total Euclidean distance

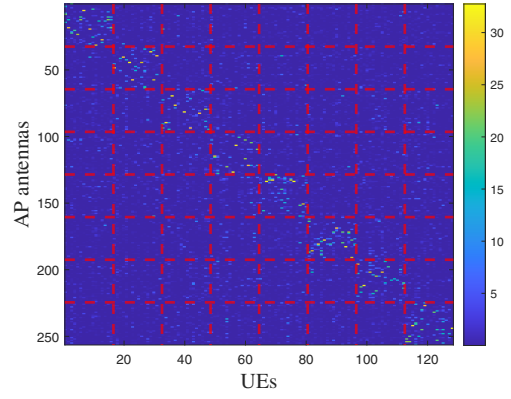


Fig. 2. The channel matrix after permutation.

between APs and their respective centroids, we can address the following optimization problem [5]:

$$\underset{\mathbf{Q} \in \mathcal{D}_{BC}}{\text{minimize}} \mathbf{1}_{1 \times B} (\mathbf{Q} \circ \mathbf{D}) \mathbf{1}_{C \times 1} - \frac{\omega}{2} \|\mathbf{Q}\|_F^2, \quad (5)$$

where

$$\begin{aligned} \tilde{\mathcal{D}}_{BC} = \{ & \mathbf{X} \in \mathbb{R}^{B \times C} : \mathbf{X} \geq 0, \\ & \mathbf{X}^T \mathbf{1}_{B \times 1} = \frac{B}{C} \mathbf{1}_{C \times 1}, \mathbf{X} \mathbf{1}_{C \times 1} = \mathbf{1}_{B \times 1} \}. \end{aligned} \quad (6)$$

Here, $\mathbf{Q}$ is the partition matrix where $Q_{b,c} = 1$ means the b -th AP is incorporated in the c -th cluster, $\mathbf{D} \in \mathbb{R}^{B \times C}$ is the distance matrix where $D_{b,c}$ indicates the Euclidean distance between the b -th AP and the centroid of the c -th cluster.

III. PROPOSED BD-GAI-BP DETECTION

A. GAI-BP Detection for CF-mMIMO

As illustrated in Fig. 3, the factor graph (FG) of BP detectors comprises two types of nodes, named respectively symbol nodes (SNs) and factor nodes (FNs). The interconnection between the two node types is determined by the characteristics of channel matrices. Notably, prior and posterior information in the form of log-likelihood ratios (LLRs) are iteratively exchanged between SNs and FNs.

In CF-mMIMO systems, APs update the posterior information and convey it to the CPU via fronthaul connections, after receiving the transmitted signals. Subsequently, the CPU transmits the prior information to the APs. Messages are exchanged iteratively between the APs and CPU until the CPU achieves an accurate enough estimation of transmitted symbols.

In this work, $\alpha_{i,j}$ denotes the messages propagated from the i -th SN to the j -th FN, which is defined as:

$$\alpha_{i,j}^{(l)}(s_k) = \log \frac{p^{(l)}(x_i = s_k)}{p^{(l)}(x_i = s_1)}, k = 1, 2, \dots, |\Theta|, \quad (7)$$

in which s_k is the k -th constellation point, Θ denotes constellation set.

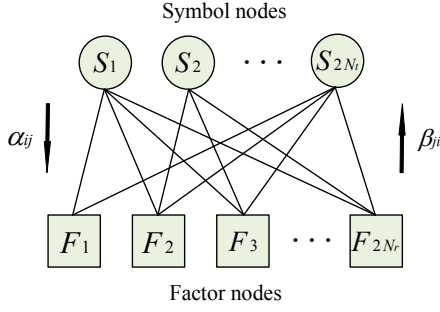


Fig. 3. The factor graph of MIMO BP detection.

Within the GAI-BP detection algorithm framework, the received signal y_j is represented as:

$$y_j = \sum_{k=1}^{2N_t} h_{j,k} x_k + n_j = h_{j,i} x_i + z_{j,i} + n_j. \quad (8)$$

The term $z_{j,i} \triangleq \sum_{k=1, k \neq i}^{2N_t} h_{j,k} x_k$ is caused by multiple-user interference (MUI), which is modeled by the *Gaussian* distribution approximately. Correspondingly, we compute the mean and variance of $z_{j,i}$ as follows:

$$\begin{cases} \mu_{z_{j,i}} = \sum_{k=1, k \neq i}^{2N_t} h_{j,k} \mathbb{E}\{x_k\}, \\ \sigma_{z_{j,i}}^2 = \sum_{k=1, k \neq i}^{2N_t} h_{j,k}^2 \text{Var}\{x_k\} + \sigma_n^2. \end{cases} \quad (9)$$

Based on the aforementioned background knowledge, the GAI-BP detection encompasses the following steps:

1) *Posterior information updating at APs*: Since the *Gaussian* approximation of MUI, the β messages sent from the j -th FN to the i -th SN can be updated by:

$$\begin{aligned} \beta_{j,i}^{(l)}(s_k) &= \frac{(y_j - \mu_{z_{j,i}}^{(l-1)} - h_{j,i} s_1)^2}{2(\sigma_{z_{j,i}}^2)^{(l-1)}} - \frac{(y_j - \mu_{z_{j,i}}^{(l-1)} - h_{j,i} s_k)^2}{2(\sigma_{z_{j,i}}^2)^{(l-1)}} \\ &= \frac{2h_{j,i}(y_j - \mu_{z_{j,i}}^{(l-1)})(s_k - s_1) - h_{j,i}^2(s_k^2 - s_1^2)}{2(\sigma_{z_{j,i}}^2)^{(l-1)}}. \end{aligned} \quad (10)$$

2) *Prior information updating at CPU*: The message $\alpha_{i,j}$ sent from the i -th SN to the j -th FN can be computed as:

$$\alpha_{i,j}^{(l)}(s_k) = \sum_{t=1, t \neq j}^{2N_r} \beta_{t,i}^{(l-1)}(s_k). \quad (11)$$

3) *Soft output computing at CPU*: After L iterations, the soft output of different symbols is given by:

$$\gamma_i(s_k) = \sum_{j=1}^{2N_r} \beta_{j,i}^{(L)}(s_k). \quad (12)$$

B. The proposed BD-GAI-BP Detection

The GAI-BP detection can achieve performance close to the theoretical limit for CF-mMIMO systems. However, it exhibits high computational complexity, rendering it unsuitable for hardware implementation in practical systems. To address this

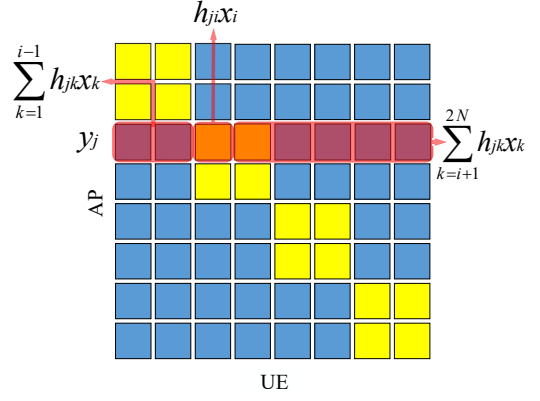


Fig. 4. The schematic diagram of proposed BD-GAI-BP algorithm.

issue, we introduce the block-diagonal GAI-BP (BD-GAI-BP) detection.

Due to the association between channel matrices and the connectivity of factor graphs, it becomes feasible to derive a partitioned factor graph based on the permuted channel matrix. The permutation operation distributes β that has a large impact on the message updating inside the corresponding blocks. We model the effect by

$$\eta_i^{(l)}(s_k) = \frac{\sum_{t \in F_i} |\beta_{ti}^{(l)}(s_k)|}{\sum_{t=1}^{2N_r} |\beta_{ti}^{(l)}(s_k)|}, k = 2, \dots, |\Theta|, \quad (13)$$

where F_i indicates the FNs belong to the same block with the i -th SN. We simplify the computation of α such that each SN only aggregates β in F_i , when $\eta_i^{(l)}(s_k)$ surpasses a specific threshold. This can be expressed as:

$$\alpha_{i,j}^{(l)}(s_k) = \sum_{t \in F_i, t \neq j} \beta_{t,i}^{(l-1)}(s_k). \quad (14)$$

The schematic diagram of the proposed BD-GAI-BP algorithm is illustrated in Fig. 4. Algorithm 1 outlines the procedural steps of the BD-GAI-BP detection. Numerical results indicate that the average value of $\eta_i^{(l)}(s_k)$ exceeds 60%. Consequently, the BD-GAI-BP detection approach effectively mitigates computational complexity without substantial performance degradation.

IV. NUMERICAL RESULTS

A. Simulation Setup

The Cell-Free channel model detailed in [5] was employed in our simulations. The 16-QAM modulation scheme was implemented with $B = N_r = 256$ and $U = N_t = 128$. Both APs and UEs were randomly distributed within a square area of 1 km². The transmission power of each UE was constrained to less than 100 mW. The carrier frequency is set to 1.9 GHz and the bandwidth to 20 MHz. Notably, the heights of the APs and UEs are respectively set at 15 m and 1.65 m. The permuted channel matrix has $C = 8$ virtual cells, therefore forming eight 32×16 blocks on the diagonal. Furthermore, the pilot sequences were assigned following the methodology proposed in [5]. All of the UEs transmitted nonorthogonal

Algorithm 1: The proposed BD-GAI-BP detection

input : received signal $\mathbf{y}$, channel matrix $\mathbf{H}$
output: the symbol soft messages $\gamma_i(s_k)$
Initialization: $\alpha^{(0)} \leftarrow \text{LMMSE}(\mathbf{y}, \mathbf{H})$;

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for  $l \leftarrow 1$  to  $L$  do
  for  $j \leftarrow 1$  to  $2N_r$  do
    for  $i \leftarrow 1$  to  $2N_t$  do
      Compute  $\mu_{z_{j,i}}^{(l)}$  and  $(\sigma_{z_{j,i}}^2)^{(l)}$  as (9);
      Compute  $\beta_{j,i}^{(l)}(s_k)$  as (10);
      Compute  $\alpha_{i,j}^{(l)}(s_k)$  as (14);
    end
  end
end
if  $l = L$  then
  Output  $\gamma_i(s_k)$  as (12);
end

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training pilots and 16-QAM payload data over $T = D = 64$ time slots, respectively. To facilitate comparison, we employed the ℓ_1 -norm-based channel estimator.

B. Performance Metrics

The UEs in CF-mMIMO systems are experiencing different SNRs, which precludes us from generating the classical BER vs. SNR plots. Therefore, we analyze the per-UE cumulative density function (CDF) for the root-mean-squared-symbol error (RMSSE) and BER. Our performance metrics are defined as follows:

1) *Per-UE RMSSE*: The RMSSE for the u -th UE over D payload data slots is defined as

$$\text{RMSSE}_u = \sqrt{\frac{\sum_{k=1}^D \left| [\hat{\mathbf{S}}_D]_{u,k} - [\mathbf{S}_D]_{u,k} \right|^2}{\sum_{k=1}^D \left| [\mathbf{S}_D]_{u,k} \right|^2}}, \quad (15)$$

where $[\hat{\mathbf{S}}_D]_{u,k}$ and $[\mathbf{S}_D]_{u,k}$ denote the estimated and transmitted data symbols of the u -th UE at time slot k , respectively. A smaller RMSSE value indicates a lower error rate. When no detection errors occur at the u -th UE, RMSSE_u converges to 0.

2) *Per-UE BER*: The BER of the u -th UE is defined as $\text{BER}_u = \frac{\varepsilon_u}{n_q}$, where ε_u indicates the average number of error bits per transmission slot, and n_q refers to the number of bits represented by each symbol.

C. RMSSE and BER Results

The simulation results of RMSSE and BER are exhibited in Fig. 5 and Fig. 6 respectively. Numerical results indicate that the GAI-BP detection supports over 99% of UEs with an RMSSE of 0.51%. In contrast, JED-FBS provides nearly 0% of UEs with such RMSSE. Additionally, GAI-BP enables nearly 99% of the UEs to achieve an uncoded BER of 10^{-3} , surpassing the corresponding ratio of JED-FBS by 28.93%. The proposed BD-GAI-BP detection algorithm supports over

93% of the UEs with an RMSSE of 0.51% and over 94% of the UEs with an uncoded BER of 10^{-3} . Moreover, these supported ratios experience only a 5.71% and 4.53% reduction compared to GAI-BP, respectively. The above numerical results demonstrate that the BD-GAI-BP achieves approximately optimal performance and surpasses the state-of-the-art algorithms.

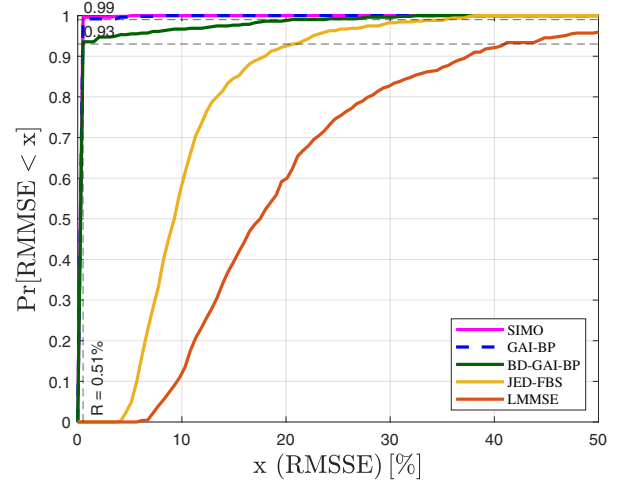


Fig. 5. The RMSSE comparison of different detectors for CF-mMIMO with $B = 256$, $U = 128$, transmitting 16-QAM symbols.

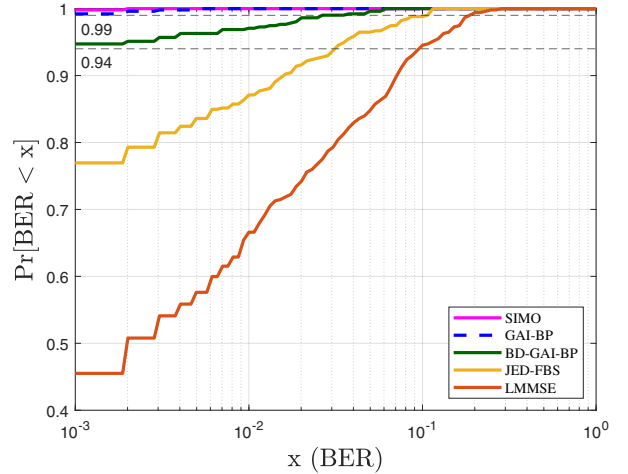


Fig. 6. The BER comparison of different detectors for CF-mMIMO with $B = 256$, $U = 128$, transmitting 16-QAM symbols.

D. Computational Complexity

In this study, the complexity of the detection algorithm is quantified based on the number of computational operations. For L iterations, the GAI-BP requires $4\sqrt{|\tilde{\Theta}|N_tN_rL}$ Mul. and Add. to update β messages, $4\sqrt{|\tilde{\Theta}|N_tN_rL}$ Add. to update α messages and $4\sqrt{|\tilde{\Theta}|N_tN_rL}$ Exp. to update prior information [10], where $\tilde{\Theta}$ denotes the set of constellation symbols in complex-domain. As there are C blocks in the permuted channel matrix, the size of F_i is $\frac{2N_r}{C}$, it reduces the

TABLE I
APPROXIMATE COMPLEXITY OF DIFFERENT DETECTION ALGORITHMS PER CHANNEL USE

Algorithm	Multiplications	Additions	Comparisons	Exponentiations
ZF/LMMSE	$\mathcal{O}(N_t^3)$	$\mathcal{O}(N_t^3)$	$\frac{N_t(N_t-1)}{2}$	0
GAI-BP	$\mathcal{O}(4\sqrt{ \tilde{\Theta} N_tN_rL})$	$\mathcal{O}(8\sqrt{ \tilde{\Theta} N_tN_rL})$	0	$\mathcal{O}(4\sqrt{ \tilde{\Theta} N_tN_rL})$
BD-GAI-BP	$\mathcal{O}(4\sqrt{ \tilde{\Theta} N_tN_rL})$	$\mathcal{O}(4(\frac{1}{C} + 1)\sqrt{ \tilde{\Theta} N_tN_rL})$	0	$\mathcal{O}(4\sqrt{ \tilde{\Theta} N_tN_rL})$

number of addition operations to $\frac{1}{C}$ of the original count in equation (14). By incorporating the updating rule for α into equation (14), the BD-GAI-BP detection effectively reduces the number of addition operations to $4(\frac{1}{C} + 1)\sqrt{|\tilde{\Theta}|N_tN_rL}$. Table I presents a comparative analysis of the complexities of different detection algorithms. Based on Table I, we assume that each addition carries half the computational complexity of multiplication, exponentiation entails twice the complexity of multiplication and the computational complexity of comparison is equivalent to that of multiplication. Numerical results illustrate that the proposed BD-GAI-BP detection achieves performance close to that of GAI-BP while saving 10.94% computational costs compared to GAI-BP. Fig. 7 provides an intuitive comparison between BD-GAI-BP and GAI-BP, which illustrates that BD-GAI-BP saves more complexity with more blocks.

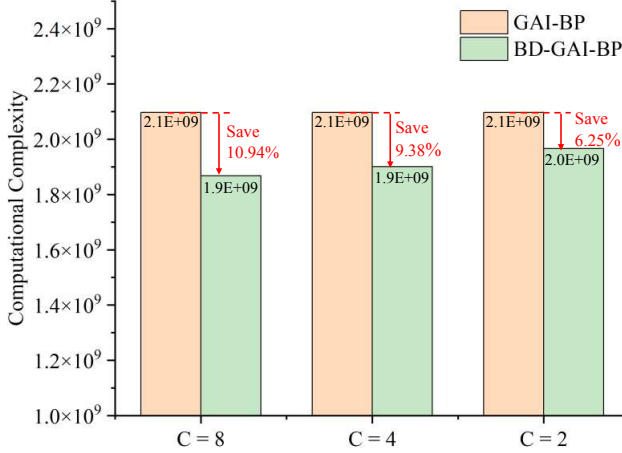


Fig. 7. The complexity comparison of GAI-BP and BD-GAI-BP for CF-mMIMO with $B = 256$, $U = 128$, transmitting 16-QAM symbols.

V. CONCLUSIONS

This paper proposed a BD-GAI-BP detection algorithm for CF-mMIMO systems. Through permutation, a block-diagonal channel matrix is acquired for the CF-mMIMO system. Exploiting the sparsity of the block-diagonal channel matrix, the proposed BD-GAI-BP detection can achieve 10.94% complexity reduction without too much performance degradation. Further work will be directed toward the efficient detection algorithm in coded CF-mMIMO.

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